

Topic 06: Tank Mixing Problems & Draining Rates

Professor Leonard Master Series · Lesson 19

Course: Applied Ordinary Differential Equations (ENGR 213 Companion)

1. Whiteboard Intuition: The Law of Conservation of Mass

Professor Leonard simplifies mixing problems down to a single undeniable physical truth:

"Rate of Accumulation = Rate of Input - Rate of Output"

$$\frac{dA}{dt} = R_{\text{in}} - R_{\text{out}}$$

Where:

- $A(t)$ is the mass of solute (pounds or kilograms) in the tank at time t .
- $R_{\text{in}} = (\text{Inflow fluid rate } r_{\text{in}}) \times (\text{Inflow concentration } c_{\text{in}})$.
- $R_{\text{out}} = (\text{Outflow fluid rate } r_{\text{out}}) \times (\text{Current tank concentration } c_{\text{out}}(t))$.
- If perfectly stirred: $c_{\text{out}}(t) = \frac{A(t)}{V(t)}$, where $V(t) = V_0 + (r_{\text{in}} - r_{\text{out}})t$.

2. Professor Leonard's Whiteboard Problem Walkthroughs

Problem 6.1: Constant Volume Mixing Tank & Asymptotic Equilibrium

Statement: A 500-gallon tank initially contains 200 gallons of brine with 30 lbs of dissolved salt. Brine containing 2 lbs of salt per gallon enters at 3 gal/min. The well-stirred mixture leaves the tank at 3 gal/min.

1. Find the amount of salt $A(t)$ in the tank at any time $t \geq 0$.
2. Find the limiting amount of salt as $t \rightarrow \infty$.

Step-by-Step Whiteboard Solution:

- **Step 1: Check Tank Volume** $V(t)$: $r_{\text{in}} = 3 \text{ gal/min}$, $r_{\text{out}} = 3 \text{ gal/min} \implies \frac{dV}{dt} = 0 \implies V(t) = 200$ gallons (constant).
- **Step 2: Calculate Inflow Rate** R_{in} :

$$R_{\text{in}} = (3 \text{ gal/min}) \times (2 \text{ lbs/gal}) = 6 \text{ lbs/min}$$

- **Step 3: Calculate Outflow Rate** R_{out} :

$$c_{\text{out}}(t) = \frac{A(t)}{200} \text{ lbs/gal}$$

$$R_{\text{out}} = (3 \text{ gal/min}) \times \left(\frac{A(t)}{200} \text{ lbs/gal} \right) = \frac{3}{200} A(t) \text{ lbs/min}$$

- **Step 4: Formulate the Differential Equation:**

$$\frac{dA}{dt} = 6 - \frac{3}{200} A \implies \frac{dA}{dt} + \frac{3}{200} A = 6$$

Initial condition: $A(0) = 30 \text{ lbs.}$

- **Step 5: Solve using Integrating Factor:**

$$\mu(t) = e^{\int \frac{3}{200} dt} = e^{\frac{3}{200} t}$$

$$\frac{d}{dt} \left[e^{\frac{3}{200} t} A \right] = 6 e^{\frac{3}{200} t}$$

$$e^{\frac{3}{200} t} A = 6 \left(\frac{200}{3} \right) e^{\frac{3}{200} t} + C = 400 e^{\frac{3}{200} t} + C$$

$$A(t) = 400 + C e^{-\frac{3}{200} t}$$

- **Step 6: Apply $A(0) = 30$:**

$$30 = 400 + C \implies C = -370$$

$$A(t) = 400 - 370 e^{-\frac{3}{200} t} \text{ lbs}$$

- **Step 7: Physical Asymptotic Limit:** As $t \rightarrow \infty$, $e^{-\frac{3}{200} t} \rightarrow 0$.

$$\lim_{t \rightarrow \infty} A(t) = 400 \text{ lbs}$$

Sanity Check

: $V_{\text{final}} \times c_{\text{in}} = 200 \text{ gal} \times 2 \text{ lbs/gal} = 400 \text{ lbs.}$ Matches physics perfectly!

Problem 6.2: Variable Volume Tank (The Overflow Catastrophe)

Statement: A large tank with a total capacity of 1000 Liters initially holds 500 L of pure water. Brine with 0.5 kg of salt per Liter enters at 4 L/min. The mixture leaves at 2 L/min.

1. When will the tank overflow?
2. What is the exact amount of salt in the tank at the moment of overflow?

Step-by-Step Whiteboard Solution:

- **Step 1: Determine the Volume Function $V(t)$:**

$$\frac{dV}{dt} = r_{\text{in}} - r_{\text{out}} = 4 - 2 = 2 \text{ L/min}$$

$$V(t) = 500 + 2t$$

- **Step 2: Find Overflow Time t_{overflow} :**

$$V(t) = 1000 \implies 500 + 2t = 1000 \implies 2t = 500 \implies t_{\text{overflow}} = 250 \text{ minutes}$$

- **Step 3: Rates of Salt:**

$$\begin{aligned} \circ R_{\text{in}} &= (4 \text{ L/min}) \times (0.5 \text{ kg/L}) = 2 \text{ kg/min.} \\ \circ R_{\text{out}} &= (2 \text{ L/min}) \times \left(\frac{A(t)}{500+2t} \text{ kg/L} \right) = \frac{2A(t)}{500+2t} = \frac{A(t)}{250+t}. \end{aligned}$$

- **Step 4: Differential Equation in Standard Form:**

$$\frac{dA}{dt} + \frac{1}{250+t}A = 2, \quad A(0) = 0$$

- **Step 5: Integrating Factor:**

$$\mu(t) = e^{\int \frac{1}{250+t} dt} = e^{\ln(250+t)} = 250 + t$$

- **Step 6: Multiply and Integrate:**

$$\begin{aligned} \frac{d}{dt}[(250+t)A] &= 2(250+t) = 500 + 2t \\ (250+t)A &= \int (500 + 2t) dt = 500t + t^2 + C \\ A(t) &= \frac{500t + t^2 + C}{250+t} \end{aligned}$$

- **Step 7: Apply $A(0) = 0$:**

$$\begin{aligned} 0 &= \frac{C}{250} \implies C = 0 \\ A(t) &= \frac{500t + t^2}{250+t} = \frac{t(500+t)}{250+t} \text{ kg} \end{aligned}$$

- **Step 8: Evaluate at $t = 250$ min (Overflow):**

$$A(250) = \frac{250(500 + 250)}{250 + 250} = \frac{250(750)}{500} = \frac{750}{2} = 375 \text{ kg}$$

Concentration at overflow

$$: \frac{375 \text{ kg}}{1000 \text{ L}} = 0.375 \text{ kg/L.}$$

Problem 6.3: Two-Tank Coupled Cascade

Statement: Tank 1 contains 100 L with 20 kg salt. Pure water flows into Tank 1 at 5 L/min. Tank 1 drains into Tank 2 (100 L, initially pure water) at 5 L/min. Tank 2 drains out at 5 L/min. Find the amount of salt $A_2(t)$ in Tank 2 at any time t .

Step-by-Step Whiteboard Solution:

- **Step 1: Solve Tank 1 First:**

$$\frac{dA_1}{dt} = 0 - 5 \left(\frac{A_1}{100} \right) = -\frac{1}{20} A_1 \implies A_1(t) = 20e^{-t/20}$$

- **Step 2: Formulate Tank 2:** Input to Tank 2 is the output of Tank 1: $R_{\text{in},2} = \frac{5}{100} A_1(t) = \frac{1}{20} (20e^{-t/20}) = e^{-t/20}$. Output of Tank 2: $R_{\text{out},2} = \frac{5}{100} A_2(t) = \frac{1}{20} A_2(t)$.

$$\frac{dA_2}{dt} + \frac{1}{20} A_2 = e^{-t/20}, \quad A_2(0) = 0$$

- **Step 3: Solve Tank 2 with Integrating Factor:**

$$\mu(t) = e^{t/20}$$

$$\frac{d}{dt} \left[e^{t/20} A_2 \right] = e^{t/20} \cdot e^{-t/20} = 1$$

$$e^{t/20} A_2 = t + C \implies A_2(t) = (t + C)e^{-t/20}$$

- **Step 4: Apply** $A_2(0) = 0 \implies C = 0$:

$$A_2(t) = te^{-t/20} \text{ kg}$$

Notice that $A_2(t)$ peaks at $t = 20$ minutes with $A_{2,\text{max}} = 20/e \approx 7.36$ kg!

Problem 6.4: Torricelli's Law for Tank Draining

Statement: A cylindrical water tank of height H and radius R drains through a small bottom hole of area a . If the water velocity exiting is $v = \sqrt{2gh}$, derive the time T required to empty the tank completely.

Step-by-Step Whiteboard Solution:

- **Step 1: Conservation of volume:**

$$dV = A_{\text{tank}}(h)dh = -avdt$$

$$\pi R^2 dh = -a\sqrt{2gh}dt$$

- **Step 2: Separate variables:**

$$h^{-1/2}dh = -\frac{a\sqrt{2g}}{\pi R^2}dt$$

- **Step 3: Integrate from** $h = H$ **at** $t = 0$ **to** $h = 0$ **at** $t = T$:

$$\int_H^0 h^{-1/2}dh = -\frac{a\sqrt{2g}}{\pi R^2} \int_0^T dt$$

$$\left[2h^{1/2} \right]_H^0 = -2\sqrt{H} = -\frac{a\sqrt{2g}}{\pi R^2}T$$

- **Step 4: Solve for T :**

$$T = \frac{2\pi R^2 \sqrt{H}}{a\sqrt{2g}} = \frac{\pi R^2}{a} \sqrt{\frac{2H}{g}}$$

3. Common Exam Traps & Professor Leonard Warnings

- **Trap 1: Assuming Volume is Constant:** When $r_{\text{in}} \neq r_{\text{out}}$, $V(t) = V_0 + (r_{\text{in}} - r_{\text{out}})t$. The denominator is NOT constant!
- **Trap 2: Forgetting Inflow Concentration:** If pure water enters, $c_{\text{in}} = 0 \implies R_{\text{in}} = 0$. If brine enters, $R_{\text{in}} = r_{\text{in}} \cdot c_{\text{in}}$.